

TrialScope: Global Clinical Trial Access Monitor

1. Team Information

- **Team Name:** Eco Hackers
- **Track Chosen:** Track 2 - Give AI Eyes
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2. Problem Statement

Approximately 80% to 85% of global clinical trials fail to meet their patient enrollment targets on time, causing multi-million dollar delays in life-saving medicine infrastructure. Major registries (e.g. ClinicalTrials.gov, EU Clinical Trials Register, WHO ICTRP) and pharmaceutical portals aggressively geo-filter trial availability listings based on the visitor's IP address.

As a result, critical cancer patients or rare-disease diagnostic seekers in developing nations like Sri Lanka are shown a highly restricted subset of active trials. Meanwhile, global pharmaceutical conglomerates desperately require South Asian patient cohorts for genetic diversity mapping but possess no automated pipelines to connect with them. This systemic information gap creates deep healthcare inequality and an institutional recruitment block within a USD 3 billion global market.

3. Proposed Solution

TrialScope is an autonomous, multi-region AI agent terminal designed to give healthcare networks and clinical trial sponsors an unblocked, cross-border viewport into global clinical recruitment ecosystems.

Core Product Features:

- **Autonomous Registry Mapping:** Multi-agent browser nodes concurrently scrape, query, and monitor global clinical registries and private pharma portals (e.g., Roche, Pfizer) across multiple target nations simultaneously.
- **Unified Eligibility Intake:** Uses a natural language processing pipeline to ingest unstructured medical conditions, parsing eligibility criteria, age constraints, and regional site allowances into a normalized global map.
- **Sponsor Cohort Alerter:** Generates automated telemetry notifications and data feeds for contract research organizations (CROs) when eligible, ethnically diverse patient cohorts register an interest.
- **Global Access Portal:** A web dashboard enabling clinical departments to instantly view missing trial opportunities that are hidden behind regional geo-filters but accept cross-border patients.

4. Why Proxies Are Needed

In clinical trial aggregation, a proxy infrastructure is the **load-bearing legal and technical validation instrument** required to see hidden medical data.

Public and private trial databases enforce strict, regulatory-bound CDN geo-fencing frameworks. For instance, the EU Clinical Trials Register serves highly specific criteria fields strictly to European Union IP connections, rendering them invisible to South Asian terminal requests. Furthermore, major pharmaceutical portals completely block traffic originating from non-Western cloud infrastructures or standard corporate VPN endpoints to mitigate web scraping.

Torch Labs Residential Proxies mapped across global target countries simulate local citizens, forcing the target web applications to reveal their localized recruitment catalogs. Without residential proxy routing, the system remains completely blind to hundreds of invisible clinical options.

5. Technical Build Plan

TrialScope operates as an asynchronous parallel data worker engine built for high-accuracy string extraction:

- **Collection Infrastructure:** Headless Playwright workers running inside Python routines, configured to dynamically pipe navigation headers through global Torch Labs proxy networks.
- **Semantic Analysis Layer:** Utilizes advanced LLM data parsing logic to clean multi-lingual medical parameters and structured PDF protocols into standardized JSON telemetry objects.
- **Relational Storage Core:** High-performance PostgreSQL backend mapping long-term spatial data changes over time.

Required Monitored System Endpoints:

- **GET /health:** Yields fundamental core controller status mapping `{"status": "ok"}`.
- **POST /config:** Programmatically modifies ingestion interval timers, lookup definitions, and timeout counters.
- **POST /proxies:** Mounts active Torch Labs credential models and defines localized geographic targeting maps.
- **GET /proxies:** Exposes live connection latencies, query success metrics, and proxy node block ratios.

Working MVP Target (By July 12): A functional single-page web console where dropping a trial diagnostic condition instantly pulls, evaluates, and overlays cross-border trial tracking variances across 5 parallel global proxy nodes.

6. Business Case

- **Target Marketplace:** Global pharmaceutical corporations (Novartis, Pfizer, Roche), Contract Research Organizations (CROs), hospital oncology units, and international medical research groups.
- **Commercialization Structure:** High-margin multi-tier enterprise SaaS framework:
 - *Institutional Tier (\$200/month):* Geared toward localized clinical departments to unlock unblocked global search pipelines and patient advisory logging tools.
 - *Pharma Strategic Tier (\$1,500+/month):* Built for global developers, providing custom portal scraping targets, automated cohort profiling tools, and full data stream API access.
- **Unit Economics:** Network proxy data usage structures scale safely with active search query routines, estimated at USD 60 to 90 monthly per active tracking cell, driving premium gross margins well above 75%.

7. Scope, Timeline, and Proxy Requirements

- **In Scope (By July 12):** Core orchestration engine loop, automated browser stealth routing layers, live multi-source lookup components across 5 critical registries, and metric analysis dashboards.
- **Explicitly Out of Scope:** Patient data handling portals, real-time doctor-facing chat integrations, and standalone mobile application distributions.
- **Technical Risk Mitigation:** Brittle target design changes are circumvented by mapping raw HTML streams straight into zero-shot text extraction layers instead of static web element queries.

Torch Labs Infrastructure Matrix & Support:

- **Proxy Categories Required:** Access to global **Residential Proxies** (critical to bypass pharmaceutical corporate web application firewalls and view geo-filtered listings).
- **Support Request Specification:** YES. We require active implementation support from the Torch Labs infrastructure team to arrange low-latency country overrides (specifically US, EU, and Asian pools) and help optimize browser cookie persistence behaviors for complex multi-page registry queries.

